

Intermodal Transfer & Connection Fees: Alaska Bulk Fuel Delivery

Blind multi-method derivation with adversarial verification
of USD per gallon modal-handoff fees, plus the storage-free
rate-aware re-derivation and worked routing examples

DOE MAS fuel-logistics routing model — cost-rate research
Chapters 5–6, extending the four per-gallon-mile transport-rate chapters
Blind tariff / operational / revealed-cost methods, skeptic panels,
local-document cross-check, and end-to-end route sanity checks

July 28, 2026

05 — Intermodal Transfer Fees

Tooling / provenance note (required disclosure): This chapter synthesizes a blind three-method derivation (tariff build-up / operational build-up / revealed-cost decomposition) for each of seven modal-handoff fees, each run past two independent adversarial skeptic panels and a local-document cross-check. Every load-bearing figure was verified verbatim against its primary source (Crowley Bethel rate sheet; Port of Bethel Tariff #5 with 2026 rate columns; Port of Nome Tariff 16.4/16.5; Port of Alaska Tariff 10.1/10.2; ISER 2010 *Components of Alaska Fuel Costs*; ATRI 2025 Update; AK DOL Pamphlet 600 Issue 52; AEA FY2025 PCE report; NSB FY26-27 budget book; NWAB Noatak Winter Fuel Haul System). All values are **2026 dollars**. Where a method was refuted by the skeptics it is discounted below; where methods diverged, the value with the strongest surviving provenance is chosen and the divergence is explained.

Architecture (fixed, not re-litigated here): the blended **Barge** per-gallon-mile rate (0.0010) already absorbs lightering + most terminal handling (revealed at \$0.8-1.2/gal on Bethel-type routes), so the fee table carries **no lightering entry**. The **Road** rate (0.0007) and **IceRoad** rate (0.010) are carriage-only, so rack/handling/receiving fees legitimately live in the fee table. Every derivation below excludes carriage, storage rent, product cost, and taxes by construction.

05.1 — ("barge", "storage"): barge discharge into hub tank farm (marine header fee)

Event: one line-haul barge discharging diesel/heating oil across a marine header into a western-Alaska hub tank farm (Bethel/Nome/Kotzebue class). Excludes carriage, lightering, storage rental, truck rack.

Tariff — \$0.17 (range 0.10-0.23). Two builds bracketing the event. Municipal wharfage: Port of Bethel Tariff #5 §206(D) inbound petroleum \$0.06/gal (2026 column); Port of Nome 16.5 \$0.048/gal (2025) → \$0.049 (2026\$). Terminal-operator header fee: Crowley Bethel rate sheet Marine Header \$0.1500/gal (eff. 8/10/2022) × CPI 1.11 → \$0.167. Cost-side floor = wharfage \$0.048-0.06 + ISER-2010 shore-side offload labor \$0.04-0.06 × 1.51 = \$0.06-0.09 + dockage ~\$0.001 → **\$0.11-0.15**. Market build = Crowley \$0.167 + wharfage where a city dock is crossed → **\$0.167-0.227**. Point = CPI-adjusted Crowley header (\$0.17).

Ops — \$0.09 (range 0.06-0.15). 2.5M-gal event, 1,500 gpm effective, ~31.8 h. Bottom-up shore-side ops (crew + monitoring + header amortization + spill readiness) ≈ \$0.032/gal + municipal wharfage \$0.055 = \$0.086. ISER anchors the high end (\$0.15) via fully-allocated 24-h receiving labor.

Revealed — \$0.15 (range 0.09-0.22). ISER cost-build (a) offload labor \$0.04-0.06 + (b) city dock fee \$0.02-0.04 (2010\$), terminal-space fee \$0.07-0.15 correctly excluded as storage. (a)+(b) × 1.51 = \$0.09-0.15. Bethel/Nome tariff cross-checks \$0.11-0.21. Point \$0.15.

Skeptics. All three methods **upheld** on source integrity — every figure located verbatim, CPI arithmetic correct within ~2%. Two corrections shared by both panels: (1) the ops **\$0.016/gal vessel-time-on-the-clock is a double-count** (port/discharge time lives in the blended barge per-mile rate) → drop it, but ops simultaneously **under-counts allocated shore labor** vs ISER's fully-loaded \$0.06-0.09, netting to ~\$0.08-0.11; (2) the revealed derivation **misreads ISER's \$0.10-0.15 "missing-header" differential** — that is a carriage-difficulty premium for header-less communities, not a discharge fee — and its additive Bethel double-stack is probably a single levy recorded twice → drop both → ~\$0.12. Panel 1 recommended **\$0.13 (0.09-0.18)**; Panel 2 recommended **\$0.11 (0.07-0.17)**.

Local cross-check. ISER is the only on-point local PDF; all cited figures verbatim. The city-dock fee (\$0.02-0.04, 2010\$) CPI-adjusts almost exactly onto today's Nome/Bethel wharfage — independent validation of both the inflation method and the tariff floor. Local reading **confirms** the skeptic corrections (missing-header differential is a transport-charge variation, not a discharge fee) and endorses **~\$0.11, range \$0.07-0.17**.

Convergence & final. The Crowley \$0.167 is a genuine market price but is the *third-party-throughput ceiling*; most hub tank farms are carrier-owned (Crowley Bethel/Kotzebue, Bonanza Nome) and pay the internal cost, not the posted margin. The verified cost build (municipal wharfage \$0.05-0.06 + fully-allocated shore-side receiving \$0.06-0.09) is \$0.11-0.15. Weighting the two skeptic panels (0.13, 0.11) and the local ISER-anchored \$0.11:

| **Final: \$0.12/gal, range \$0.07-0.17 (2026\$), medium confidence.**

05.2 — ("storage", "overland"): tank-farm loading rack onto tanker truck (truck rack fee)

Event: one truck-rack loading event at a hub bulk terminal (rack use, metering, loading labor, VCU). Excludes storage rental, marine intake, carriage.

Tariff — \$0.25 (range 0.01-0.30). Exact-match anchor: Crowley Bethel Table 2 Truck Rack \$0.2500/gal (eff. 8/10/2022), rack-only by construction (storage \$2.25/bbl-mo and marine header \$0.15 are separate line items). $\text{CPI} \times 1.114 \rightarrow \0.279 (2026\$). ISER third-party terminal fee \$0.07-0.15 (2010\$) $\times 1.51 = \$0.11$ -0.23 brackets just below. Urban low-end (Valero MTSA, AIAS flowage) ~\$0.01. Rural-hub-dominated network → point \$0.25, shaded just below the CPI-adjusted Crowley figure.

Ops — \$0.12 (range 0.02-0.28). 50/50 urban/rural blend of an engineering build-up (rural bottom-up \$0.13 vs Crowley cross-check \$0.28 → rural rep ~\$0.20; urban ~\$0.03).

Revealed — \$0.21 (range 0.05-0.32). Crowley \$0.28 (rural) 2/3-weighted with a state-heating-oil-contract urban intercept (~\$0.06) 1/3-weighted → \$0.207.

Skeptics. Both panels **upheld the tariff method** (Crowley figure verbatim-verified, storage separately billed → genuinely rack-only, no carriage double-count) and **refuted ops and revealed:** ops' \$0.12 rests on unsourced rack-capital/throughput assumptions and a 50/50 blend whose urban leg is fabricated (no observed Alaska road-system rack fee); revealed's urban intercept is an unfounded decomposition of a bundled state-contract delivered price plus a mis-scoped airport *flowage tax*. Corrected toward the sole real anchor, both collapse to ~\$0.20-0.25. Shared dollar-year nit: actual June-2026 CPI-U is 327.075 (not 330-334), so CPI-correct Crowley is \$0.276 not \$0.282 (immaterial). Panel 1 recommended **\$0.22 (0.03-0.30)**; Panel 2 recommended **\$0.24 (0.10-0.30)**.

Local cross-check. ISER has no truck-rack-specific line — only terminal-*access* (\$0.07-0.15) and terminal-*use* (\$0.03), both broader than a pure rack event. CPI-corrected ($\times 1.500$) that is \$0.105-0.225 and \$0.045. The derived point sits at/above the top of the access band, defensible *only* because it rests on the external Crowley rack tariff. Local recommends **modest downward shading to ~\$0.20**, range low to ~\$0.03-0.045, and strongly endorses bimodality (~\$0.03-0.06 road-system racks vs ~\$0.20-0.25 remote barge hubs).

Convergence & final. One verified exact-match Alaska tariff (Crowley \$0.276 CPI-2026\$) anchors everything; the debate is only how far to shade for embedded sole-provider rent (ISER's cost-structure "terminal use" is $\sim 8\times$ lower). Panels landed 0.22-0.24; local ISER pulls to ~0.20. Splitting:

Final: \$0.22/gal, range \$0.05-0.30 (2026\$), medium confidence. Strongly bimodal — if the model can key on hub class, use ~\$0.05 for road-system racks and ~\$0.25 for remote Western-Alaska barge hubs rather than one blended value.

05.3 — ("barge", "overland"): direct barge-to-truck handoff at a landing

Two readings must be kept separate, because the model uses this key two ways.

(a) Component-sum through hub storage (the model's default construction). This key = barge→storage + storage→overland. With the retuned legs, **\$0.12 + \$0.22 = \$0.34/gal (range 0.12-0.47)**.

(b) Genuine direct barge-to-truck at a landing (no tank transit). A dedicated derivation priced this event:

- **Tariff — \$0.12 (0.05-0.22):** landing/dock fee (Port of Alaska Item 260 fuel-transfer 1.90¢/gal 2026; Bethel \$200.E \$0.06/gal; ISER city-dock \$0.03-0.06) ~\$0.05 + shore-side offload labor (ISER \$0.04-0.06 $\times 1.5$ = \$0.06-0.09) ~\$0.07.
- **Ops — \$0.16 (0.07-0.40) and Revealed — \$0.22 (0.12-0.40):** both dominated by barge-standby / "no-terminal slow-discharge premium" components.

Skeptics. Both panels **refuted ops and revealed** and **upheld only the tariff build**. The decisive fault is scope: ops re-charges $\sim 2/3$ of its number as barge standby-hours, and revealed rebuilds ISER's missing-header (\$0.10-0.15) and deficiency (+\$0.20) *delivered-price variations* — both are barge

carriage/dwell already smeared into the 0.0010 barge per-mile rate. Panel 1 (stripping only the double-count) converged **\$0.12 (0.06-0.20)**; Panel 2 (also noting the model already charges \$0.40 here and the barge rate carries offload labor) converged **\$0.08 (0.04-0.14)**.

Local cross-check. ISER substantiates both critiques verbatim: the \$0.10-0.15 figure is a transportation-charge variation (p.23), the barge's own carriage is fully allocated at \$0.45/gal (Table 1), and Table 4 assigns only \$0.03 to discrete "terminal use." Local recommends **~\$0.10, range \$0.05-0.15** for the direct event, explicitly *not* carrying the \$0.22-0.40 highs.

Convergence & final. For the **component-sum construction the model actually uses**, the honest value is the sum of the two retuned legs:

Final (component path): \$0.34/gal, range \$0.12-0.47 (2026\$), medium confidence. If this key ever fires on a true direct landing (no storage): use ~\$0.10/gal, range \$0.06-0.20 (surviving tariff build; ops/revealed refuted for double-counting barge dwell).

05.4 — ("plane", "overland"): air-cargo fuel offload / tarmac pumping

Event: ~40-min engines-off tarmac offload of a DC-6/C-46 bulk-fuel load (2,000-4,000 gal, 3,000 divisor) into a truck or village tanks. Excludes air carriage, landing fees, avgas.

Tariff — \$0.12 (0.05-0.25). No standalone tarmac tariff exists; Everts sells charter-bundled. Build: 3-person crew self-offload (Everts crews load/unload their own aircraft) ~\$463/hr × 0.65 h ÷ 3,000 = \$0.100; onboard pump/hose amortization \$0.012; representative AIAS non-aviation flowage \$0.01 → \$0.122.

Ops — \$0.15 (0.04-0.42). Same crew + truck standby + **\$350/hr aircraft ground-standby** (\$262.50, 59% of the mid case) + equipment → \$0.149.

Revealed — \$0.13 (0.06-0.25). Heating-oil-contract intercept allocation + bottom-up (\$0.105) + marine analogues, plus a probability-weighted **\$0.02 flowage adder**.

Skeptics. Both panels **upheld tariff** (least double-count-exposed — omits airframe standby) and **refuted ops and revealed**. Ops' \$350/hr aircraft ground-standby is already inside the all-in air per-gal-mile rate, and the cited charter contract proves the first 2 h of load/unload standby are *free/bundled* → strip it → ops ~\$0.06. Revealed's flowage adder mis-applies the DOT&PF fee (it is on airport fuel *sales*/dispensing, heating fuel *exempt*, not a cargo transfer) → strip it → revealed ~\$0.10; its marine terminal analogues are the wrong physical event. Both panels recommended **~\$0.09 (0.05-0.16)**.

Local cross-check. ISER contains **no air-tarmac figure at all**; it is entirely marine/barge + trucking. Critical provenance defect flagged: the "\$1.25/gal per 100 air-miles / DC-6 4,900 gal" cross-check attributed to this local PDF **is not in it** (it belongs to a separate 2008 ISER report). The one comparable datum — shore-side offload *labor* \$0.04-0.06/gal (p.18) — sits at the low-mid of the

derived range and loosely corroborates it. Local: keep ~\$0.09, range \$0.05-0.16; remove the mis-cited air-carriage anchor.

Convergence & final. The double-count-clean labor+equipment core is \$0.06-0.11 across all three; add-ons (airframe standby, flowage) were refuted.

Final: \$0.10/gal, range \$0.05-0.16 (2026\$), low confidence (~85% engineering build-up; no published tarmac tariff). If the air per-gal-mile rate is confirmed built from all-in charter pricing, the marginal billed offload trends to \$0.05-0.06.

05.5 — ("plane", "drums"): manual offload of drummed fuel from aircraft

Event: manual offload of 55-gal fuel drums from a small aircraft at a village strip. Scope explicitly **excludes** the drum packaging as a commodity input (retail layer) and includes only handling-cycle drum attrition.

Tariff — \$0.85 (0.45-1.50). Labor (Pamphlet 600 A0710 Material Handler \$52.66/hr, 1.25 person-h ÷ 165 gal) \$0.40 + equipment \$0.04 + **handling-cycle drum attrition** (10%/cycle × ~\$130 delivered) \$0.24 + spill/hazmat \$0.18 = \$0.86.

Ops — \$0.80 (0.45-2.00). Same structure but drum term = **full new-drum capital** (\$110 ÷ 4 fills = \$150/event = \$0.545/gal, 66% of the point).

Revealed — \$0.60 (0.35-1.10). PCE decomposition: Takotna \$5.63 vs supply hub McGrath \$4.96 → \$0.68 gap; net air carriage & origin load-out → ~\$0.33 destination residual + drum amortization → ~\$0.58-0.65.

Skeptics. The scope-decisive issue is whether new-drum capital is in-scope. **Both panels refuted ops** (its dominant term is full drum capital, which the model's separate commodity/retail layer carries; its own caveat concedes the correction to \$0.25-0.30). **Tariff and revealed upheld** — both scope the drum term to attrition/loss. The two panels then diverge on the residual: **Panel 1** (attrition-*included* as legitimate handling damage) → **\$0.62 (0.40-1.10)**; **Panel 2** (strip *all* drum capital, keep only offload-cycle damage) → **\$0.35 (0.20-0.65)**.

Local cross-check. ISER has no drum figure; its closest analog (marine 24-h linehaul offload labor \$0.04-0.06/gal over 250,000+ gal) is 10-20× lower purely because volume-per-event is enormous — which confirms the *premise* (per-gallon handling explodes at 110-275-gal loads) without independently pinning magnitude. Its spill/admin benchmark (\$0.05-0.10/gal, volume-spread) modestly favors the stricter Panel 2's lean spill line over the tariff's \$0.18. Local leans toward ~\$0.35 (0.20-0.65) but calls it a lean, not a mandate.

Convergence & final. The two scope-clean skeptic centers straddle \$0.35 (drum-capital-excluded) and \$0.62 (attrition-included). The current \$0.50 placeholder sits squarely between them and inside

both ranges; with real provenance now behind it, it is a defensible single value pending a model decision on the drum-commodity layer.

Final: \$0.50/gal, range \$0.20-1.10 (2026\$), low confidence. Conditional guidance: if the model prices drums in a separate commodity layer, use ~\$0.35; if this fee is meant to carry drum replacement, use ~\$0.62. Per-gallon cost scales inversely with shipment size (single-drum C207 loads push toward the high end).

05.6 — ("overland", "ice_road"): highway tanker to ice-road truck via hub storage

Event: offload a highway tanker into Dalton/Deadhorse hub storage + reload a ~3,000-gal ice-road tanker (two passes). Excludes highway and ice-road carriage.

Tariff — \$0.16 (0.10-0.25). ISER item-3 third-party terminal (through-)fee \$0.07-0.15 (2010\$) $\times$ 1.50 = \$0.105-0.225, midpoint \$0.165; small-lot reload uplift and truck-cheaper-than-marine-intake offsets net ~0. POA Item 260 1.90¢/gal and Nome wharfage bracket.

Ops — \$0.10 (0.05-0.20). Bottom-up: Pamphlet 600 A2101 (incl. "Fueller/Oil Distributor Truck") \$84.34/hr fully-loaded + ATRI standby \$20.17/hr + pump rates $\rightarrow$ Step A (8,500-gal offload) \$0.035 + Step B (3,000-gal reload) \$0.067 = \$0.102.

Revealed — \$0.30 (0.16-0.45). Anchors on Crowley Bethel in+out (\$0.15 marine header + \$0.25 rack) $\times$ CPI $\rightarrow$ \$0.28-0.45, picking \$0.30 on a "sole-provider high-cost like Bethel" premium.

Skeptics. Both panels **refuted revealed**: its intake leg is a **marine header** applied to a **highway-tanker** intake (double-prices offload the road rate already covers), and its monopoly premium is wrong for a road-connected hub with multiple operators (Colville, Crowley, Inlet) — its own bottom-up cross-check lands \$0.14-0.19. **Tariff upheld** but leans slightly high (stacks a small-lot uplift on a range that already prices full terminal use) $\rightarrow$ ~\$0.14. **Ops upheld** as the cleanest build (best-sourced labor) but is a marginal-cost floor missing O&M/capital/margin $\rightarrow$ grossed up ~\$0.14. Panel 1 recommended **\$0.12 (0.08-0.20)**; Panel 2 recommended **\$0.17 (0.10-0.30)**.

Local cross-check. NSB FY26-27 budget confirms a real in-house Deadhorse/North-Slope hub operator (FNG division, NSB-owned tank farms, Deadhorse HBTC) — supporting the *cost-basis* framing over the monopoly-margin marine tariff, i.e. favoring the downward correction of revealed. Noatak PDF confirms the store-then-reload chain and 3,000-5,000-gal parcel. No per-gallon rate in any local doc, so no numeric move.

Convergence & final. Corrected estimates converge \$0.09 (ops) / \$0.14 (tariff) / \$0.18 (revealed-adjusted); panels centered 0.12-0.17.

Final: \$0.14/gal, range \$0.08-0.25 (2026\$), medium confidence. High end accommodates a genuine Deadhorse camp/per-diem premium + small-lot reload; low end

removes any hub-O&M carried elsewhere in the model.

05.7 — ("ice_road", "storage"): ice-road truck discharge into village tank farm

Event: single discharge of an ice-road tanker/cat-train (3,000-5,000 gal) into a village tank farm — one pass. Excludes ice-road carriage, tank-farm O&M.

Tariff — \$0.22 (0.10-0.35). Nome wharfage \$0.049 + 2-person × 2.0-h receiving labor (Nome \$115/\$180 blended \$156/hr) \$0.156 + equipment \$0.011 = \$0.216. Crowley rack \$0.28 as upper anchor.

Ops — \$0.07 (0.04-0.16). Bottom-up: 1.75-h event, cold-derated pump, driver-operator + village operator + ATRI equipment standing → \$279/event ÷ 4,000 gal = \$0.070.

Revealed — \$0.15 (0.08-0.25). PCE: Atqasuk \$7.41 vs four barge-direct NSB comparators mean \$5.71 → \$1.70 winter-haul premium; net terminal + carriage, time-proportional discharge share (8.2%) → \$0.111 + village receiving \$0.034 = \$0.144.

Skeptics. Both panels **refuted the tariff method**: its anchors (Crowley \$0.25 rack, Nome wharfage) are the wrong unit of service — a monopoly terminal-throughput fee + a marine wharfage applied by assumed symmetry to a truck event — and ISER Table-4 "terminal use" (\$0.03/gal) makes a \$0.22 bare-discharge handoff implausibly large; its 2-person × 2.0-h labor is the unsupported 71%-of-total driver. **Ops upheld** (\$0.07, verified inputs). **Revealed upheld** but biased high by un-stripped storage/admin → nudge to ~\$0.10. Both panels stress the **delivering crew/rig time is already inside the 0.010 ice-road carriage rate** (which per `blind_derivations/04_ice_road.md` includes a +20% load/unload adder over the full round-trip cycle); **only village-side receiving labor is additive**. Panel 1 recommended **\$0.08 (0.05-0.15)**; Panel 2 (pure carriage-scope lens) recommended **\$0.05 (0.03-0.09)**.

Local cross-check. No local doc prices the discharge event. Two citation corrections: NWFHLS capital is **\$425,000** (not the \$350,000 both ops/revealed cited — a \$75k sub-line error, immaterial to their soft cross-check); and the "200 trips / 350-400k gal" figures the second skeptic panel called fabricated **are in fact present on Noatak p.10** — that skeptic point is itself inaccurate. NSB village Fuel cost centers (Atqasuk 6108.FUEL \$472,913; Nuiqsut 6110.FUEL \$569,184) are in-house staffing with no per-gallon line, supporting the ops cost-basis framing. No numeric move.

Convergence & final. Ops (\$0.07) and revealed (\$0.10, biased high) converge once the tariff outlier is corrected down; the binding architectural fact is that delivering-side handling is already in the ice-road per-mile rate, leaving only village receiving.

Final: \$0.08/gal, range \$0.04-0.15 (2026\$), medium confidence. Dominant sensitivities: load size (3,000 vs 5,000 gal) and whether a dedicated village receiving operator is required (driver self-discharge → floor).

05.8 — ("barge","ice_road"): never-traversed placeholder edge

Status (verified 2026-07-23): a same-month barge↔ice-road handoff is temporally impossible (ice roads Jan-Mar, barges May-Oct; zero month-overlap at all 8 North Slope sites), so monthly Dijkstra routing **never traverses this edge**. It exists only to satisfy the no-costless-edge rule. The real flow is inter-temporal through hub storage: barge→storage in summer, storage→ice-road in winter.

Composition. Mirror the two true component legs with their retuned values: barge→storage **\$0.12** (summer intake) + a storage-side ice-road **loading** leg (the winter half of the store-then-reload chain, ~\$0.08-0.14 per \$05.7's one-pass framing) → component sum ~**\$0.20**.

Final: \$0.20/gal, range \$0.11-0.32 (2026\$), low confidence. Because the edge is never exercised, the exact value is immaterial to results; it is set to the honest component sum rather than the stale \$0.40. Keep the `marine_header_fee` + loading sub-key structure so a future inventory-dynamics phase (Phase 6) can decompose it into the two temporal legs.

Summary table

Fee key	Current	Final (2026\$)	Range	Confidence	Change
("barge","storage")	0.15	0.12	0.07-0.17	medium	lower — Crowley 0.15 is a third-party price w/ margin; carrier-owned hubs pay the cost
("storage","overland")	0.25	0.22	0.05-0.30	medium	lower — sole-provider rack tariff shaded for monopoly rent; bimodal by hub class
("barge","overland")	0.40	0.34	0.12-0.47	medium	lower — mechanical sum of the two retuned legs (0.12+0.22)
("plane","overland")	0.157	0.10	0.05-0.16	low	lower — standby/flowage add-ons refuted; labor+equipment core ~0.09
("plane","drums")	0.50	0.50	0.20-1.10	low	keep — placeholder now provenance-backed; sits between the two scope readings (0.35/0.62)
("overland","ice_road")	0.25	0.14	0.08-0.25	medium	lower — 0.25 was a borrowed rack analog; two-pass truck handoff costs less
("ice_road","storage")	0.25	0.08	0.04-0.15	medium	lower — one pass, not two; delivering-side time already in ice-road per-mile rate
("barge","ice_road")	0.40	0.20	0.11-0.32	low	lower — never traversed; honest component sum of two retuned legs

Carry-forward limitations (values still resting on placeholder- or thin-grade evidence)

1. **("plane","drums") — scope-unresolved (low).** The point value depends on a model decision not yet made: whether new-drum capital lives in a separate commodity/retail layer (→ ~\$0.35) or is folded into this fee (→ ~\$0.62). \$0.50 is a compromise. The two dominant quantities (drum fills-before-loss; per-gallon handling time on 110-275-gal loads) have **no documented Alaska value**. Resolve by (a) fixing the drum-commodity-layer convention in the model and (b) obtaining an Everts/Ryan drum-handling or deposit figure.
2. **("plane","overland") — ~85% engineering build-up (low).** No carrier or airport publishes a standalone tarmac fuel-offload tariff; the point rests on crew-time and pump-rate estimates. **Provenance fix required:** the ISER "\$1.25/gal/100mi air-carriage" cross-check is mis-cited to the local ISER 2010 PDF — that figure is not in that document (belongs to the separate 2008 ISER *Components of Delivered Fuel Prices* report). Re-attribute or drop. Firm up with an awarded bid schedule under state Air Cargo contract 240000071 (Everts Additional Standby Time \$/hr).
3. **("ice_road","storage") and ("overland","ice_road") — no published Deadhorse/North-Slope tariff (medium).** North Slope hub terminals (Colville, Brice, Crowley Deadhorse) are private and publish no throughput fees; both values are bottom-up builds transposed from ISER marine-hub and Bethel commercial rates, cross-checked against Pamphlet 600 labor and ATRI equipment. A single confirmed operator invoice for either handoff would collapse the range. The ice_road→storage value also depends on the still-implicit assumption (from 04_ice_road.md) that the 0.010 per-mile rate fully absorbs delivering-side discharge dwell — if that rate is ever recalibrated without the load/unload adder, this fee must rise to re-capture delivering-side time.
4. **("barge","ice_road") — never exercised (low, but immaterial).** Correct only up to the accuracy of its two mirrored legs; harmless because Dijkstra cannot traverse it. Revisit only when Phase 6 inventory dynamics make the inter-temporal barge→storage→ice_road path explicit.
5. **All values — CPI to mid-2026 (minor).** Several derivations used a projected mid-2026 CPI-U of ~326-334; the actual June-2026 index is 327.075. The discrepancy moves any point estimate by <\$0.006/gal and is below the rounding of every final value here. ISER 2010 figures (the backbone of the marine-hub and receiving-labor anchors) are 16 years old; CPI-U plausibly *understates* rural-Alaska labor/insurance/spill-compliance escalation, biasing the ISER-anchored floors slightly low.
6. **Bimodality not yet keyed (("storage","overland")).** The single blended \$0.22 mis-prices both ends of a genuinely bimodal fee (~\$0.05 road-system racks vs ~\$0.25 remote barge hubs). If the network can distinguish hub class, split this fee rather than carry one value.

06 — Storage-Free Modal Connection Costs

All figures in 2026 dollars. This chapter re-derives the four modal-boundary fuel CONNECTION costs after the storage pseudo-mode was removed from the fee table. Under the option-(a) architecture each fee is now a single atomic modal-boundary crossing cost — no intake+storage-rest+outbound chain. The per-mile Barge/Plane/IceRoad rates are ALL-IN (each carries its own side's handling); the Road rate is CARRIAGE-ONLY (ATRI Tables 8/9 verified: no rack/terminal/loading line). Each fee therefore prices ONLY handling that has no home in a per-mile rate — in practice the road-carriage-side out-loading/receiving, or (where no road mode is incident) a thin facility-interface residual.

Two blind derivations were produced per boundary (ops_direct operational build-up; rate_residual rate-aware residual), then an adversarial skeptic hunted for double-counts and drop-counts. Final values are weighted by what survived the skeptic: strip any double-count, restore any drop-count.

6.1 — Barge <-> Road tanker-truck connection (("barge", "overland") , 205 Transfer edges)

Method 1 — ops_direct (\$0.025/gal, range \$0.015-0.045). Bottom-up marginal build on a 3,500-gal local hub load: 0.75-hr occupancy (14-min pump at 250 gpm + 30-min setup/teardown) x 1.5 crew x ~\$55/hr fully-loaded Alaska labor + \$20/hr standby equipment + consumables = ~\$82/load = \$0.023/gal. ISER cross-check (\$0.04-0.06/gal 2010 shore-side offload, CPI x1.53, road-slice ~1/3-1/2) brackets it. A pure marginal FLOOR; omits rack metering/capital/overhead.

Method 2 — rate_residual (\$0.28/gal, range \$0.18-0.33). Crowley Bethel Terminal tariff (eff. 8/10/2022): Marine Header \$0.15 + Truck Rack \$0.25 = \$0.40 two-sided. Barge all-in rate already carries the \$0.15 marine header -> SUBTRACT (anti-double-count); road carriage-only carries none of the \$0.25 rack -> KEEP (anti-drop-count). Residual = \$0.25 x CPI 1.10 = \$0.28/gal.

Skeptic outcome. Both derivations UPHELD; neither commits the double-count. The DEPLOYED \$0.40 total does — its \$0.15 marine header is already in the all-in barge rate (0.0010) and must be stripped. No drop-count: road-side rack handling correctly retained (ATRI has no rack line, so it has no other home). rate_residual is the stronger construction (exact-match Alaska tariff); ops_direct's \$0.025 is a floor ~10x below every posted anchor. Point shaded from \$0.28 toward low-\$0.20s for embedded sole-provider margin.

- **Counted:** road-carriage-side out-loading only (positioning, connect/bond, pumped transfer into truck, gauge/meter/paperwork, disconnect, shore attendant).
 - **Excluded:** barge-side marine header/lightering/tank-farm intake (in all-in barge rate); ISER \$0.20 deficiency adder; storage rest.
 - **Final value: \$0.24/gal, range \$0.15-0.30, high confidence.** Change: \$0.40 -> \$0.24 by stripping the double-counted \$0.15 marine header and lightly shading the \$0.25 rack.
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6.2 — Barge <-> Ice-road truck connection (("barge", "ice_road"), 8 Transfer edges, North Slope)

Method 1 — ops_direct (\$0.011/gal, range \$0.00-0.021). Full pump-through build = ~\$0.054/gal (0.94 hr x 2 crew x \$84.34/hr NSB fully-loaded + \$60/hr equipment / 4,000-gal parcel). Architecture gate: barge intake is in the all-in barge rate; ice-road load dwell is in the ice_road +20% adder — EXCLUDE both physical legs -> strict floor \$0.00. Thin genuinely-uncounted residual = cross-modal changeover coordination (0.5 operator-hr / 4,000 gal) = \$0.011/gal.

Method 2 — rate_residual (\$0.010/gal, range \$0.005-0.019). Full both-sides physical stack ~\$0.08-0.12/gal is dominated by barge intake (\$0.061-0.092 CPI-adjusted ISER) and ice-road load (\$0.02-0.03) — BOTH already in their all-in rates, subtract them, netting near-zero. Only defensible residual = the standalone vessel<->land-tanker facility interface: Port of Alaska Item 260 (pipeline transfer 0.145 c/gal floor to vessel<->truck fuel 1.90 c/gal ceiling), mid ~\$0.010/gal.

Skeptic outcome. Both UPHELD, clean — no double-count, no drop-count. Dominant finding: this is a boundary where BOTH incident modes are all-in and there is NO carriage-only road mode, so the drop-count trap cannot occur and both physical legs are correctly excluded. All sources verified exactly (NSB \$105,047 + 67% burden; ISER \$0.04-0.06; Noatak 3,000-5,000 gal; POA Tariff 10.1 2026 column). Converged \$0.011/gal.

- **Counted:** thin cross-modal changeover / facility interface handling only (gauging, sampling, bond/ground verification, pump-tender supervision).
- **Excluded:** barge intake (all-in barge rate); ice-road load/unload dwell (+20% adder); storage rest. No road rack leg exists here.
- **Final value: \$0.011/gal, range \$0.00-0.02, low confidence.** Change: \$0.40 placeholder -> \$0.011. The old \$0.40 was inherited from the barge<->overland row and double-counted both all-in legs. Completeness-only edge — never traversed same-month (barge May-Oct vs ice road Jan-Mar); near-zero routing impact.

6.3 — Road tanker <-> Ice-road truck connection (("overland", "ice_road"), latent boundary)

Method 1 — ops_direct (\$0.022/gal, range \$0.013-0.034). Continuous truck-to-truck pumped transfer of a 3,000-gal batch: ~\$112.50 both-sides operation (1.0 boundary labor-hr x \$90/hr fully-loaded North Slope + \$30/hr transfer equipment). Bill only the road (carriage-only) side ~0.55 share = \$61.9/3,000 gal = \$0.021 (2024\$) x CPI 1.05 = \$0.022/gal. Drops the ice-road-side pump-in dwell (in the +20% adder).

Method 2 — rate_residual (\$0.03/gal, range \$0.02-0.045). Top-down: ISER \$0.04-0.06/gal (2010) x1.47 CPI, halved to one side = \$0.03-0.044. Bottom-up check: ATRI \$90.89/hr -> ~\$97/hr 2026 x ~0.5 hr dwell + FERC terminal \$0.006/gal ≈ \$0.02/gal. Point \$0.03.

Skeptic outcome. ops_direct UPHELD (cleaner). rate_residual NOT UPHELD: its top-down ISER \$0.04-0.06/gal anchor is a MARINE BARGE 24-hr linehaul offload into a tank farm — barge intake handling already in the all-in barge rate, and a storage-node cost — mis-applied as a truck-to-truck header analog and inflated. Halving it does not transmute a barge cost into a road cost. STRIP the top-down; rate_residual's own clean bottom-up (~\$0.02) then agrees with ops_direct. The \$0.03 point survived only by two offsetting errors canceling. Converged on ~\$0.022.

- **Counted:** road-carriage-side share of the metered PTO pump-out + connect/disconnect, bond/ground, cold-weather transfer equipment (not the carriage truck).
 - **Excluded:** ice-road-side load/unload dwell (+20% adder); both trucks' carriage economics; storage rest (the atomic crossing, not truck->tank->truck).
 - **Final value: \$0.022/gal, range \$0.014-0.032, medium confidence.** Change: \$0.25 -> \$0.022. The old \$0.25 was an explicit storage/tank-rack analog (two-leg chain, TODO seed); removing the storage rest and stripping the double-counted ISER marine anchor lands the atomic fee at ~\$0.02. Latent Dalton-corridor boundary.
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6.4 — Air-cargo fuel <-> Road tanker connection ("plane", "overLand") , latent boundary)

Method 1 — ops_direct (\$0.03/gal, range \$0.015-0.10). Tarmac hose transfer of a 3,000-gal parcel: road-side receiver labor 0.83 hr x \$60/hr fully-loaded Alaska tanker CDL = \$50 + road-side equipment (receiving tanker standby, PTO pump, hoses, grounding) ~\$37 = \$87/3,000 gal = \$0.029/gal. Excludes the aircraft's own onboard pump, crew, and ground standby (all in the all-in charter rate).

Method 2 — rate_residual (\$0.038/gal, range \$0.03-0.05). ISER \$0.04-0.06/gal (2010) as full both-sides handling; subtract ~50% aircraft side (in charter rate); road residual \$0.02-0.03 x CPI 1.53 = \$0.038/gal.

Skeptic outcome. ops_direct UPHELD. rate_residual NOT UPHELD: its ISER anchor is a barge linehaul offload INTO A TANK FARM — barge intake (already in the all-in barge rate) plus storage-node receiving labor — imported into an air<->truck fee; the 50% subtraction avoids the plane double-count but the residual still inherits the barge/tank-farm basis, over-stating road-only handling. Shaded down. ops_direct's road-side PTO-pump equipment leg is also softly over-attributed because Everts DC-6/C-46 carry their own onboard offload pumps (pumping already in the charter rate); trim the equipment leg. Converged on ~\$0.025.

- **Counted:** road/ground-side receiving handling only — receiver/driver connect-monitor-disconnect labor (bond/ground, camlock, pump watch, breakaway) + modest fittings/grounding-gear charge (NOT an independent pumping engine).
- **Excluded:** aircraft onboard pump, flight crew, aircraft ground standby/positioning (all in the all-in plane charter rate 0.025); storage rest.

- **Final value: \$0.025/gal, range \$0.015-0.045, medium confidence.** Change: \$0.157 -> \$0.025 by removing aircraft-side handling (already in the plane rate) and the storage rest, and shading the barge/tank-farm-proxy anchor. Latent/low-frequency (air fuel economical only <~5,000 gal within a few hundred air-miles of a refinery).

6.5 — Summary table (2026\$)

Boundary	Old through-storage	New storage-free fee	Range	Confidence	Counts
("barge", "overland")	0.40	0.24	0.15-0.30	high	road-carriage-side rack out-loading only
("barge", "ice_road")	0.40	0.011	0.00-0.02	low	thin cross-modal changeover / facility interface only (no road mode incident)
("overland", "ice_road")	0.25	0.022	0.014-0.032	medium	road-carriage-side share of truck-to-truck transfer
("plane", "overland")	0.157	0.025	0.015-0.045	medium	road/ground-side receiving handling only

Recurring construction rule: bill ONLY handling with no home in a per-mile rate. Because Road is carriage-only, that is the road-carriage side of every crossing; the all-in side (barge intake, aircraft pump/crew, ice-road +20% dwell) is excluded to prevent double-counting; storage rest is excluded entirely. Where no road mode is incident (barge<->ice_road), only a thin facility-interface residual remains, so the fee collapses near zero.

6.6 — Coupling to per-mile rates

The four per-gallon-mile transport rates and their includes-basis are UNCHANGED by removing the storage leg. The rates remain:

- **Barge 0.0010 — all-in.** Carries vessel-to-shore lightering, marine header, and marine-terminal/tank-farm intake handling.
- **Plane 0.025 — all-in charter.** Carries the aircraft's onboard offload pump, flight/ground crew during offload, and aircraft positioning/standby.
- **IceRoad 0.010 — all-in.** Carries an explicit +20% load/unload adder over the round-trip cycle (both origin fill and destination discharge dwell).
- **Road 0.0007 — carriage-only.** Fuel, tractor/trailer, R&M, tires, insurance, permits, driver wages over running miles. ATRI Tables 8/9 verified: NO rack/terminal/loading/unloading line.

The storage pseudo-mode that was deleted never lived inside any per-mile rate — it was an artifact of the old through-storage fee construction (intake + storage-rest + outbound). Re-scoping the fees to the

atomic crossing moves nothing into or out of a per-mile rate; the change is confined to the fee table. Consequently:

- **All-in-side handling stays home in its all-in rate.** Marine intake in Barge; aircraft handling in Plane; ice-road load/unload dwell in IceRoad. This is exactly why those legs are EXCLUDED from the connection fees (double-count guard). The deployed \$0.40 barge<->overland total violated this by re-billing the \$0.15 marine header — corrected here to \$0.24.
- **Road-carriage-side handling is the only genuinely-homeless handling, and it correctly lives in the connection fees.** Because Road is carriage-only, into-truck pumping, connect/disconnect, bond/ground, gauging/metering, and receiver dwell have no home in the Road rate. Every fee is scoped to exactly this (drop-count guard).
- **Hub storage is priced nowhere** — not in a per-mile rate, not in a boundary fee. It was never a node; if delivered-cost storage is later needed it must be added as a separate component, never folded into a rate or a crossing fee.

Net: no per-mile rate changes, no orphaned handling under option (a). The all-in side handling sits in its all-in rate; the road-carriage-side handling sits in the connection fee; storage is intentionally out of scope.

6.7 — Worked routing examples (sanity check)

To confirm the storage-free fees do not distort mode choice, six illustrative routes were priced end to end. **Distances are hypothetical but Alaska-plausible**; the point is the *ranking* between modes, not the exact dollar value. Route cost (\$/gal) = $\Sigma(\text{leg miles} \times \text{per-mile rate}) + \Sigma(\text{connection fee at each mode change})$, using the per-mile rates from Chapter 6.6 and the new connection fees from the summary table above.

Route	Carriage	Handoff	Total	(old fees)
A1 — Barge to a river village (1320 mi barge + 40 mi truck)	1.348	0.240	1.588	1.748
A2 — Fly that same village instead (400 mi plane + 5 mi truck)	10.004	0.025	10.029	10.161
B1 — Winter ice road to a North Slope village (500 mi truck + 68 mi ice road)	1.030	0.022	1.052	1.280
B2 — Fly that village in winter instead (300 mi plane + 3 mi truck)	7.502	0.025	7.527	7.659
C1 — Straight road delivery, no handoff (600 mi truck)	0.420	0.000	0.420	0.420
C2 — Same town by barge + short truck (1100 mi barge + 15 mi truck)	1.111	0.240	1.351	1.510

Three checks, all pass:

1. **The cheap-to-expensive order held.** In every head-to-head the expected mode still wins, and by a wide margin: barge beats plane $\sim 6\times$ (A1 vs A2), ice road beats plane $\sim 7\times$ (B1 vs B2),

and road beats barge where a road exists (C1 vs C2, \$0.42 vs \$1.35). Plane remains the last resort; road remains cheapest.

2. **The lowered handoffs do not over-promote plane or ice road.** A plane costs \$0.025 *per mile*, so even a short flight dwarfs its now-\$0.025 handoff. The per-mile rates are far enough apart (plane 25×, ice road 14×, barge 1.4× the road rate) that shaving a connection fee never flips a mode choice over any realistic distance.

3. **The handoff fee matters only where it should.** On barge routes it is a real slice (~15% of the A1 total), because loading fuel off a barge onto a truck at a hub is genuine work. On the long, expensive plane and ice-road hauls it is a 2–3% rounding error. Correct shape.

Net effect of the re-derivation: every total fell modestly versus the old through-storage fees (right column), but **no ranking flipped** — old and new agree on which mode wins every comparison. The correction adjusted the *level* of cost without changing any *decision*, which is the expected signature of a fix that only removed double-charging. The one value that looks anomalous in isolation — `barge↔ice_road` at ~\$0.01 — never appears in a real route, since barges (May–Oct) and ice roads (Jan–Mar) never share a month.